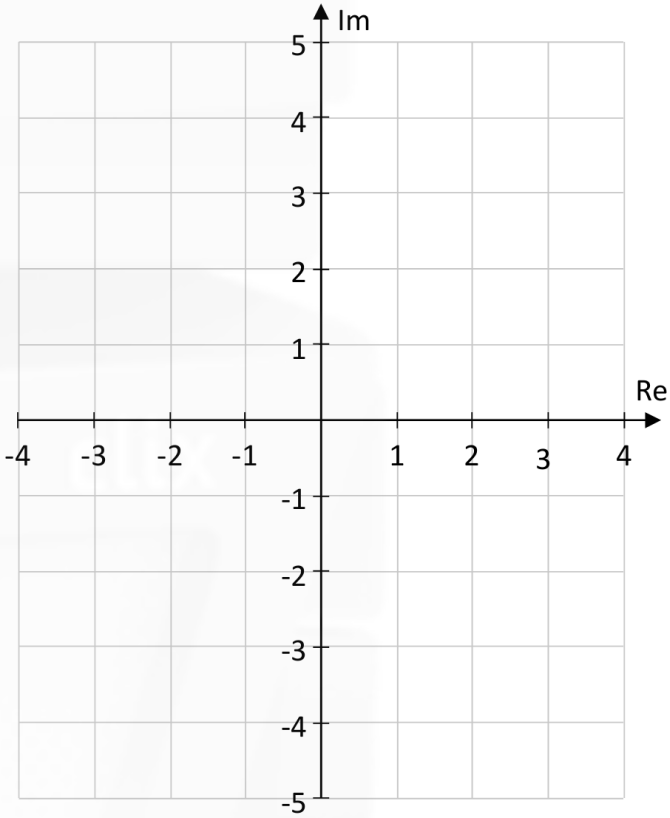
Question 3 **(25 marks)**

$z_1 = 3 - 4i,\ z_2 = -2 + i$ and $z_3 = 2iz_2$, where $i^2 = -1$.

(a) **(i)** Write z_3 in the form $a + bi$, where $a, b \in \mathbb{Z}$.

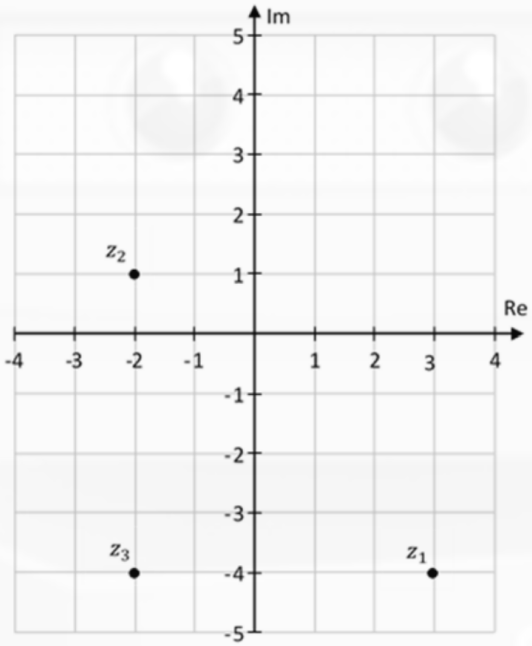
$z_3 = 2iz_2 =$

(ii) Plot z_1, z_2 and z_3 on the given Argand Diagram.
Label each point clearly.



(iii) Find $|z_1|$.

(b) If $z_1 \times z_4 = 29 + 3i$, write z_4 in the form $a + bi$, where $a, b \in \mathbb{R}$.

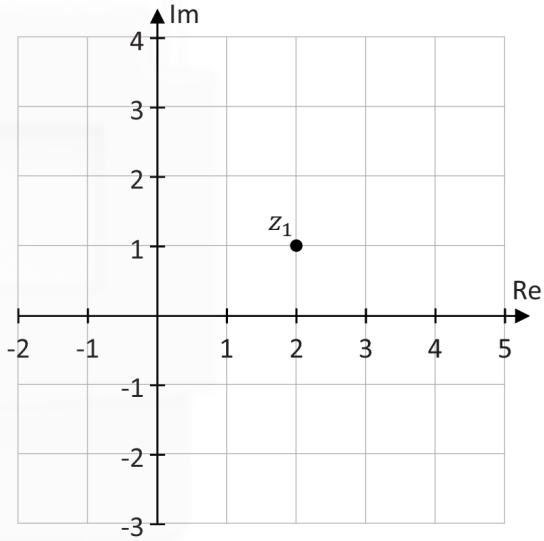
Q3	Model Solution – 25 Marks	Marking Notes
(a) (i) + (a) (ii)	$z_3 = 2i(-2+i) = -4i + 2i^2$ $-2-4i$  	**Scale10D (0, 3, 5, 8, 10)** **Note**: 4 items (find z_3 and 3 plots) *Low Partial Credit:* One correct item (correct plot for incorrect z_3) or (z_1 plotted correctly) or (z_2 plotted correctly) or (z_3 correct but with incorrect plot) *Mid Partial Credit:* 2 correct items *High Partial Credit:* 3 correct items
(a) (iii)	$\lvert z_1 \rvert = \sqrt{9+16} = 5$	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* $\lvert z_1 \rvert$ formula *High Partial Credit:* Formula fully substituted
(b)	$z_4 = \dfrac{(29+3i)(3+4i)}{(3-4i)(3+4i)}$ $= \dfrac{87+116i+9i+12i^2}{9-16i^2}$ $= \dfrac{75+125i}{25}$ $= 3+5i$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* Some correct substitution Conjugate identified Some multiplication above and below by same number, even if incorrect conjugate *High Partial Credit:* $\dfrac{(29+3i)(3+4i)}{(3-4i)(3+4i)}$

Question 2 **(25 marks)**

The complex number $z_1 = 2 + i$, where $i^2 = -1$, is shown on the Argand Diagram below.

(a) **(i)** $z_2 = 2z_1$.

Find the value of z_2, and **plot and label** it on the Argand Diagram.

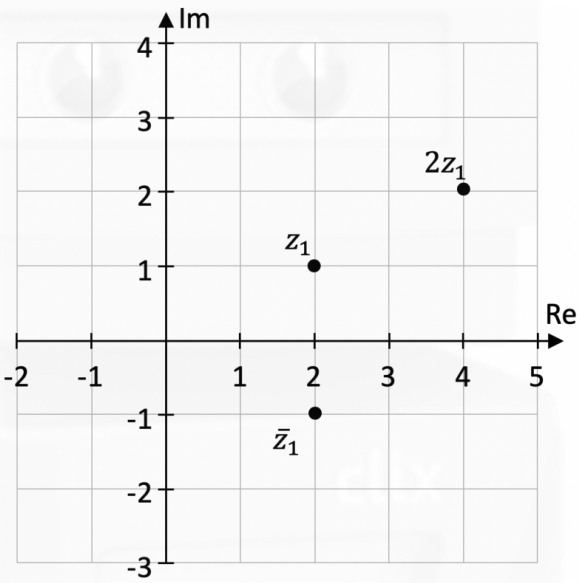


(ii) $\bar{z}_1$ is the complex conjugate of z_1.
Write down the value of $\bar{z}_1$, and **plot and label** it on the Argand Diagram.

$\bar{z}_1 =$

(iii) Investigate if $|z_2| = |z_1 + \bar{z}_1|$.

(b) Show that $z_1 = 2 + i$ is a solution of the equation $z^2 - 4z + 5 = 0$.

(a) (i) + (ii)	$z_2 = 2(2+i) = 4+2i$ $\bar{z}_1 = 2 - i$  	**Scale10D (0, 3, 5, 8, 10)** *Low Partial Credit:* • Correct substitution into $2z_1$ • Finds z_2 or $\bar{z}_1$ correctly and stops *Mid Partial Credit:* • Finds $4+2i$ and $2-i$ • z_2 or $\bar{z}_1$ found correctly and plots, with or without label *High Partial Credit:* • z_2 and $\bar{z}_1$ found and at least one plotted correctly, without labels • Mixes up axes but finds, plots and labels correctly **Note:** Ignore extra incorrect plotting
(a) (iii)	$\lvert z_2 \rvert = \sqrt{16+4} = \sqrt{20}$ $\lvert z_1 + \bar{z}_1 \rvert = 4 \neq \sqrt{20}$	**Scale 5D (0, 2, 3, 4, 5)** *Low Partial Credit:* • Correct formula • Some correct substitution • Finds $z_1 + \bar{z}_1$ *Mid Partial Credit:* • Finds $\lvert z_2 \rvert$ or $\lvert z_1 + \bar{z}_1 \rvert$ *High Partial Credit:* • Finds $\lvert z_2 \rvert$ and $\lvert z_1 + \bar{z}_1 \rvert$ but no conclusion or incorrect conclusion

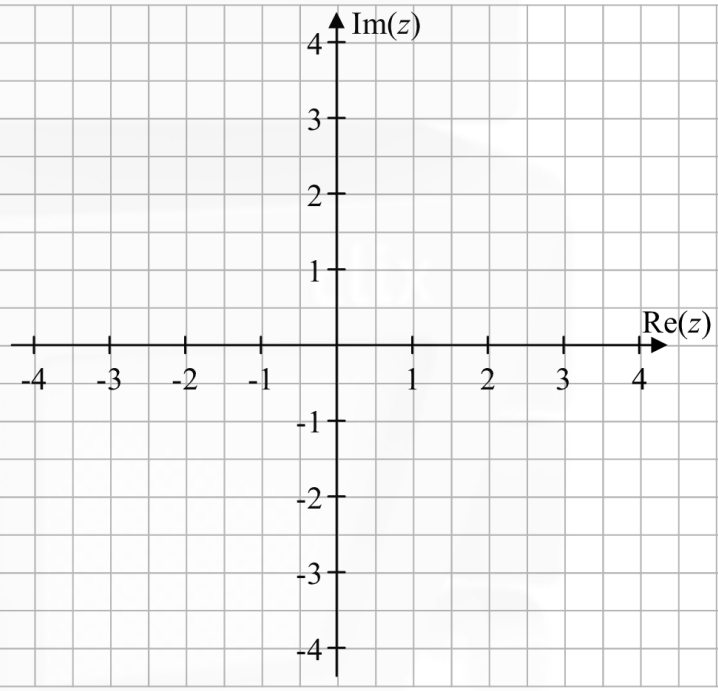
(b)	$(2+i)^2 - 4(2+i) + 5 = 0$ $4 + 4i + i^2 - 8 - 4i + 5 = 0$ $9 - 9 + 4i - 4i = 0$ or $\frac{4 \pm \sqrt{16 - 4(1)5}}{2}$ $= \frac{4 \pm \sqrt{-4}}{2}$ $= 2 \pm \mathrm{i}$	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • Some relevant substitution *High Partial Credit:* • Full substitution with correct multiplication of $(2+i)^2$ or $-4(2+i)$ **Note**: failure to draw a relevant conclusion from incorrect work to be treated as an error or **Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • Some relevant substitution into quadratic roots formula • Quadratic roots formula written *High Partial Credit:* • Full substitution with some errors but with complex roots and relevant conclusion given • Gets $\frac{4 \pm \sqrt{-4}}{2}$ **Note**: failure to draw a relevant conclusion from incorrect work to be treated as an error

Question 3 **(25 marks)**

Two complex numbers are $u = 3 + 2i$ and $v = -1 + i$, where $i^2 = -1$.

(a) Given that $w = u - v - 2$, evaluate w.

(b) Plot u, v, and w on the Argand diagram below.



(c) Find $\dfrac{2u + v}{w}$.

Question 3 **(25 marks)**

Two complex numbers are $u = 3+2i$ and $v = -1+i$ where $i^2 = -1$.

(a) given that $w = u - v - 2$ evaluate w.

We calculate this by grouping the real and imaginary parts of w together:

$$w = (3+2i) - (-1+i) - 2 = 3+2i+1-i-2 = 2+i$$

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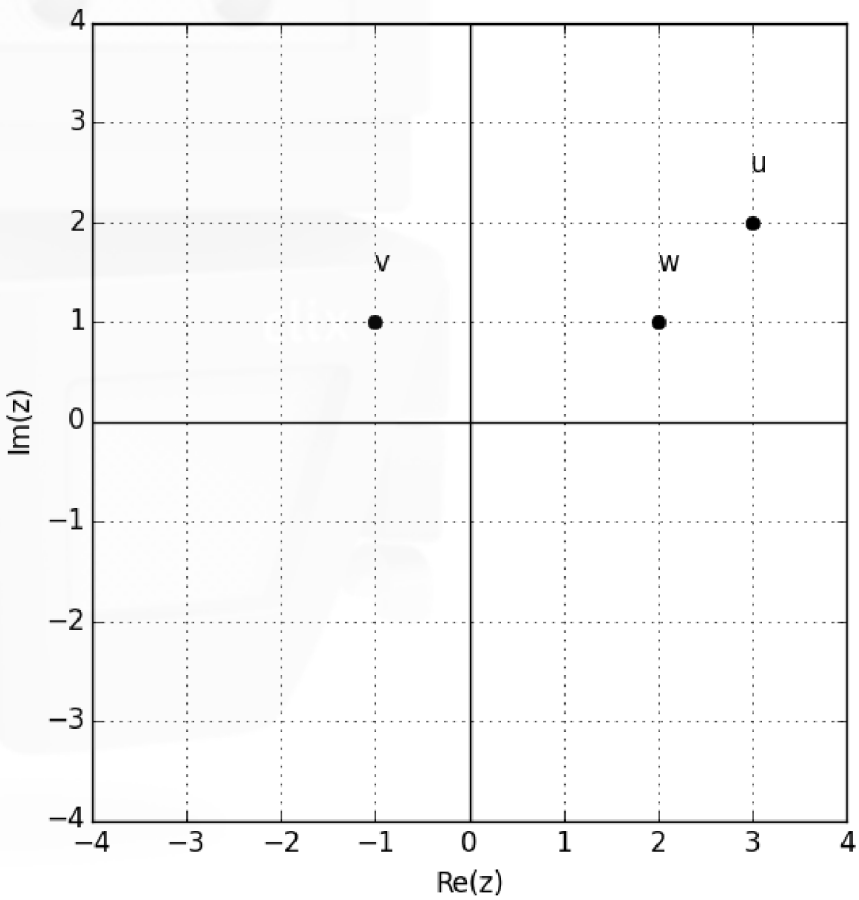
(b) Plot u, v and w on the Argand diagram.

We can plot points on the Argand diagram by considering the real and imaginary parts of the complex number as x and y coordinates respectively.



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(c) Find $\dfrac{2u+v}{w}$.

Firstly, we'll simplify the top line. $2u+v=2(3+2i)+(-1+i)=6+4i-1+i=5+5i$. To divide these two complex numbers, we'll multiply the fraction top and bottom by the conjugate of the denominator, w.

$$\begin{aligned}
\frac{2u+v}{w} &= \frac{5+5i}{2+i} \\
&= \frac{(5+5i)(2-i)}{(2+i)(2-i)} \\
&= \frac{10-5i+10i-5(i^2)}{4-2i+2i-(i^2)} \\
&= \frac{15+5i}{5} \\
&= 3+i
\end{aligned}$$

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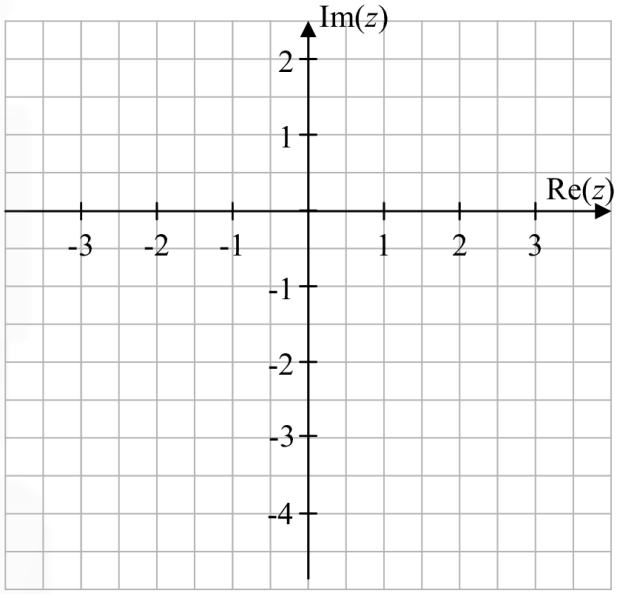
Let $z_1 = 3 - 4i$ and $z_2 = 1 + 2i$, where $i^2 = -1$.

(a) Plot z_1 and z_2 on the Argand diagram over.



(b) From your diagram, is it possible to say that $|z_1| > |z_2|$?
Give the reason for your answer.

Answer:

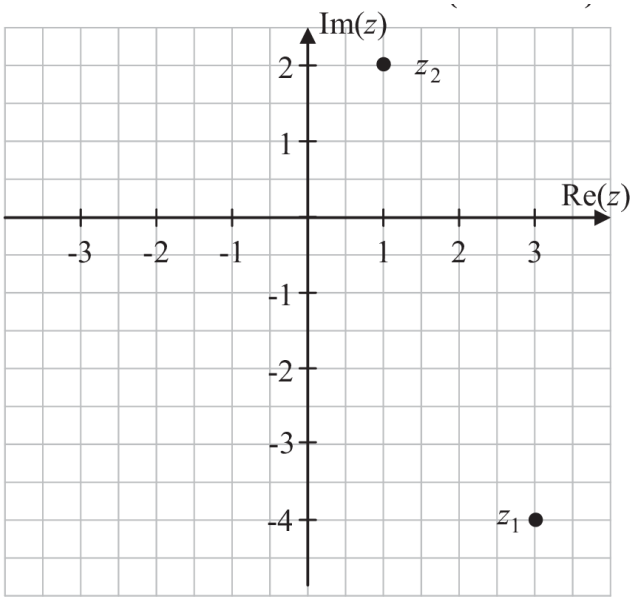
Reason:

(c) Verify algebraically that $|z_1| > |z_2|$.

(d) Find $\dfrac{z_1}{z_2}$ in the form $x + yi$, where $x, y \in \mathbb{R}$.

Let $z_1 = 3 - 4i$ and $z_2 = 1 + 2i$, where $i^2 = -1$.

(a) Plot z_1 and z_2 on the Argand diagram over.



(b) From your diagram, is it possible to say that $|z_1| > |z_2|$?
Give the reason for your answer.

Answer: Yes

Reason: The distance from the origin to z_1 is greater than the distance from the origin to z_2.

$$|z_1| = |3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$
$$|z_2| = |1 + 2i| = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$$
$$5 > \sqrt{5} \quad \Rightarrow \quad |z_1| > |z_2|$$

(d) Find $\dfrac{z_1}{z_2}$ in the form $x + yi$, where $x, y \in \mathbb{R}$.

$$\frac{z_1}{z_2} = \frac{3 - 4i}{1 + 2i} = \frac{3 - 4i}{1 + 2i} \times \frac{1 - 2i}{1 - 2i} = \frac{3 - 6i - 4i + 8i^2}{1^2 + 2^2} = \frac{-5 - 10i}{5} = -1 - 2i$$

Question 3 **(25 marks)**

(a) Write each of the following complex numbers in the form $a+bi$, where $i^2=-1$.

$z_1=(3+2i)(2-5i)=$

$z_2=(5+4i)(17-13i)-(5+3i)(17-13i)=$

$z_3=\left(\frac{5}{2}+\frac{7}{2}i\right)^2-\left(\frac{5}{2}+\frac{1}{2}i\right)^2=$

$z_4=1+i+i^2+i^3=$

(b) Which of z_1 and z_2 above is farther from 0 on an Argand diagram?
Justify your answer.

Question 3 **(25 marks)**

(a) Write each of the following complex numbers in the form $a+bi$, where $i^2=-1$.

(i) $z_1=(3+2i)(2-5i)$

$$z_1=(3+2i)(2-5i)=6-15i+4i+10=16-11i$$

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(ii) $z_2=(5+4i)(17-13i)-(5+3i)(17-13i)$

$$\begin{aligned} z_2 &= (5+4i)(17-13i)-(5+3i)(17-13i) \\ &= [(5+4i)-(5+3i)](17-13i) \\ &= i(17-13i) \\ &= 13+17i \end{aligned}$$

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(iii) $z_3 = \left(\frac{5}{2}+\frac{7}{2}i\right)^2 - \left(\frac{5}{2}+\frac{1}{2}i\right)^2$

$$\begin{aligned} z_3 &= \left(\frac{5}{2}+\frac{7}{2}i\right)^2 - \left(\frac{5}{2}+\frac{1}{2}i\right)^2 \\ &= \left(\frac{25}{4}+\frac{35}{2}i-\frac{49}{4}\right) - \left(\frac{25}{4}+\frac{5}{2}i-\frac{1}{4}\right) \\ &= 0+15i-12 \\ &= -12+15i \end{aligned}$$

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(iv) $z_4 = 1+i+i^2+i^3$

$$z_4 = 1+i+i^2+i^3 = 1+i+(-1)+(-i) = 0+0i$$

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(b) Which of z_1 and z_2 above is farther from 0 on an Argand diagram? Justify your answer.

Answer: z_2 is farther from 0.

Justification: The distance of a complex number from 0 is known as the modulus. Hence, the complex number with the greatest modulus will be the one which is farthest from 0. Calculating these we find that z_2 is farther than z_1 from 0:

$$|z_1| = |16-11i| = \sqrt{(16)^2+(-11)^2} = \sqrt{377}$$
$$|z_2| = |13+17i| = \sqrt{(13)^2+(17)^2} = \sqrt{458}$$

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